Research Abstract

Dyslexia is a neurodevelopmental learning disorder that affects accurate word recognition, spelling, decoding, and phonological processing, impacting approximately 5–10% of children worldwide. Beyond academic challenges, children with dyslexia often experience reduced self-confidence, anxiety, fear of failure, and difficulties in social interaction. Despite the rapid growth of educational technologies, most existing digital learning tools for dyslexia are limited to single-player environments that primarily focus on decoding accuracy and provide feedback only after errors occur. These approaches fail to address the broader cognitive, emotional, and social needs of dyslexic learners, and often reinforce negative learning experiences through repetitive error-based correction mechanisms.

This research introduces a novel **multimodal multiplayer learning framework** designed to support both cognitive and social skill development in children with dyslexia. The proposed system integrates real-time collaborative gameplay within an immersive virtual environment developed using Unreal Engine, combined with a mobile companion application built using Flutter to enable cross-device interaction. The system architecture leverages WebSocket-based communication and multiplayer networking to synchronize gameplay, enabling multiple users to engage in cooperative literacy tasks such as word formation, letter recognition, and problem-solving activities in real time. Additionally, a voice-enabled communication system is incorporated to facilitate peer interaction in a safe and anonymous virtual environment, helping children overcome communication anxiety and build social confidence.

A key innovation of this system is the implementation of a **non-competitive reward-based motivation model**, which replaces traditional leaderboard-based competition with achievement-driven progression, positive reinforcement, and personalized feedback. This approach is designed to reduce performance pressure and create an emotionally supportive learning environment. Furthermore, the system incorporates **adaptive personalization mechanisms** that dynamically adjust task difficulty, feedback intensity, and learning pathways based on individual user performance and cognitive load, ensuring that each learner progresses at an optimal pace without experiencing frustration or disengagement.

Another significant contribution of this research is the development of an **anticipatory multimodal feedback system**, which provides guidance before errors occur using a combination of visual cues (highlighting, animations), auditory prompts (phoneme-based sounds, instructions), and haptic feedback (vibrations). This proactive feedback mechanism contrasts with traditional post-error correction methods and aims to support error-free learning, improve accuracy, and enhance retention. The mobile application also

includes interactive modules such as letter tracing, color selection, and drawing-based tasks, which contribute to motor skill development and multisensory engagement, further reinforcing learning outcomes.

To evaluate the effectiveness of the proposed system, a controlled experimental study was conducted with 30 dyslexic children aged 7–12. Participants were exposed to both single-player and multiplayer learning conditions, and multiple performance metrics were collected, including task completion accuracy, engagement levels (time spent, attempts), cognitive load (measured using adapted NASA-TLX), and social interaction indicators. The system also recorded real-time gameplay data such as error rates, retries, and collaboration patterns to support quantitative analysis.

The results demonstrate that the multimodal multiplayer approach significantly improves learning outcomes compared to traditional single-player systems. Specifically, the system achieved over 98% task completion accuracy, increased user engagement, reduced cognitive load, and enhanced social presence and peer interaction. Participants reported higher motivation, reduced anxiety, and improved confidence in both learning and communication tasks. These findings validate the core hypothesis that integrating multiplayer collaboration, adaptive personalization, and anticipatory feedback can create a more effective and inclusive learning environment for children with dyslexia.

In conclusion, this research contributes a comprehensive, scalable, and innovative framework for next-generation dyslexia learning systems by bridging the gap between cognitive training and social development. The proposed solution not only enhances literacy skills but also addresses the emotional and social barriers faced by dyslexic learners. The outcomes of this study provide valuable design guidelines for inclusive educational technologies and highlight the potential for expanding such systems into broader applications, including virtual reality environments, large-scale educational platforms, and global accessibility-focused EdTech solutions.